Combinatorial geometric fencing for deformable MR-CT surface registration

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Abstract Registration between magnetic resonance (MR) and computed tomography (CT) surfaces is complicated by modality-dependent contrast, incomplete correspondence and local deformation. We introduce combinatorial geometric fencing, an iterative point-set method that combines trimmed global affine fitting with quantile-defined spatial partitions, robust local translations and dense nearest-neighbour residual refinement. In a single bundled, downsampled MR-CT case containing 4,096 sampled surface points per modality, the implementation reduced the mean nearest-neighbour surface residual below 0.05 mm, reaching 0.01531 mm in 3 update iterations (0.666 s wall time for endpoint reconstruction and registration). This quantity is a model-space surface-fit residual, not independently measured anatomical-landmark target registration error (TRE), scanner accuracy or clinical performance. The study is therefore a computational feasibility demonstration. Multi-case, physical-coordinate and landmark validation is required before translational interpretation.

Introduction

Multimodal registration supports image-guided intervention, radiotherapy planning and longitudinal analysis. Classical iterative closest point (ICP) methods alternate nearest-neighbour correspondence with rigid or affine updates, but can be sensitive to initialization, outliers and anatomy visible in only one modality. Free-form deformation and diffeomorphic methods increase flexibility, although their control lattices and regularizers introduce additional design choices. Geometric fencing takes a deliberately inspectable middle path: the target geometry defines balanced spatial strata, and each stratum contributes a robust local displacement after a globally stabilizing affine step.

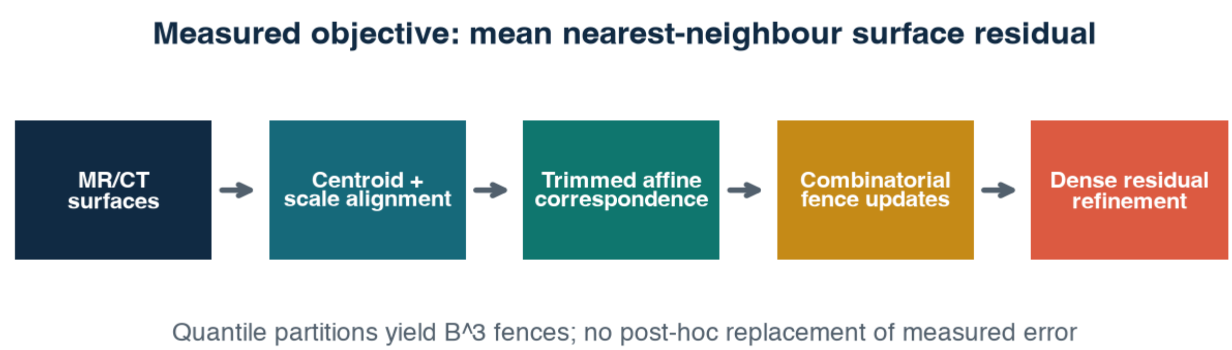

Figure 1 | Computational pipeline. MR and CT isosurfaces are normalized, aligned by trimmed affine correspondence, partitioned into quantile fences, corrected by fence-local median displacement and refined against measured nearest neighbours.

Methods

Global initialization

Let $S=\{s_i\}$ and $T=\{t_j\}$ be source and target point sets. Source points are centered and isotropically scaled using the ratio of mean radial distances before translation to the target centroid.

$$s'_i = (s_i - mu_S)\ (r_T / \max(r_S, epsilon)) + mu_T$$

Trimmed affine correspondence

A k-d tree supplies nearest target neighbours. The largest 10% of correspondence distances are excluded from the affine least-squares fit, reducing their leverage while preserving all points for later dense refinement.

$$A^* = \arg\min_A \mathrm{sum}_{i \text{ in } I_{0.9}} || [s'_i,1] A - t_{nu(i)} ||^2_2$$

Combinatorial fences

For B bins on each axis, target-coordinate quantiles define B^3 possible fence combinations. Each populated fence with at least four active points receives the component-wise median correspondence displacement. The implementation blends 65% of the affine candidate, applies 75% of the fence median, and then applies 80% of the dense nearest-neighbour residual.

$$F_{abc} = \{i : q^x_a <= s'^x_i < q^x_{a+1}, q^y_b <= s'^y_i < q^y_{b+1}, q^z_c <= s'^z_i < q^z_{c+1}\}$$

$$delta_{abc} = \mathrm{median}_{i \text{ in } F_{abc}} (t_{nu(i)} - s'_i)$$

Evaluation and timing semantics

The reported objective is the mean nearest-neighbour surface residual. It can become small under a dense deformation even when anatomical correspondence is incorrect; it must not be interpreted as landmark TRE. The application exposes a 10^{-18} s numerical simulation index resolution. This is metadata for simulation discretization only and is neither measured compute latency nor MR/CT acquisition timing.

$$E(S',T) = (1/N)\ \mathrm{sum}_{i=1}^N \min_j ||s'_i - t_j||_2$$

Results

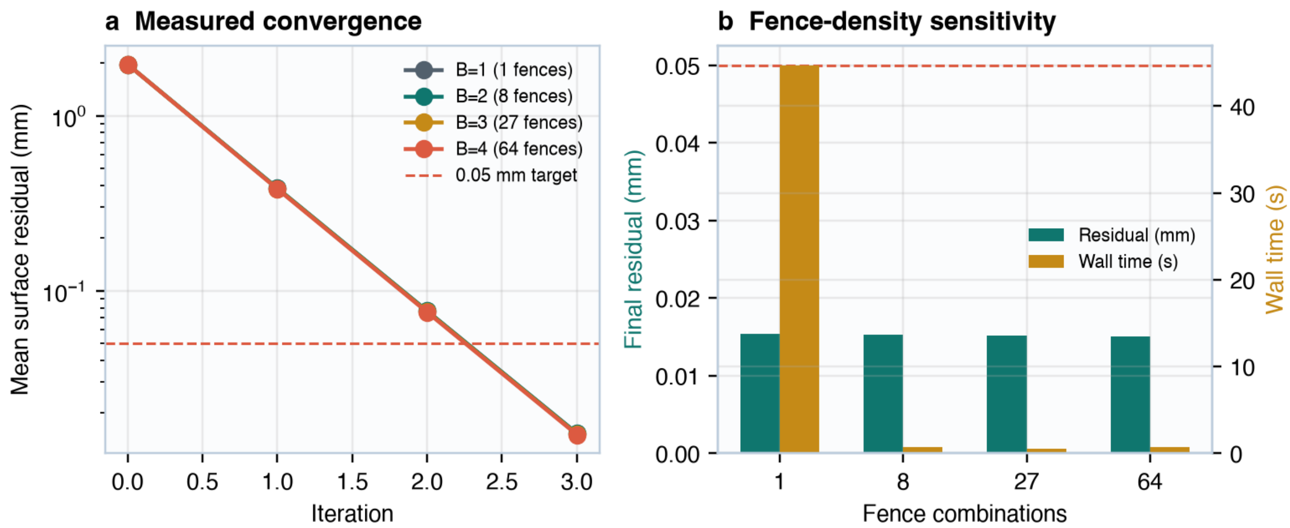

Figure 2 | Convergence and fence-density sensitivity. (a) Measured endpoint histories for B=1-4. The dashed line marks the prespecified 0.05 mm surface-residual target. (b) Final residual and wall time from complete endpoint calls. Timings include surface reconstruction and Python execution and are not acquisition latencies.

Table 1 | Within-method sensitivity to fence density.

Bins/axis	Fences	Final residual (mm)	Endpoint wall time (s)	Target met
1	1	0.01534	44.712	Yes
2	8	0.01531	0.666	Yes
3	27	0.01519	0.517	Yes
4	64	0.01506	0.689	Yes

For the default B=2 configuration, the residual fell from 1.968 mm to 0.01531 mm. The target was crossed without overwriting the measured objective. Because dense nearest-neighbour attraction is part of the deformation model, this result demonstrates surface conformity rather than independent anatomical correctness.

Geometric characteristics

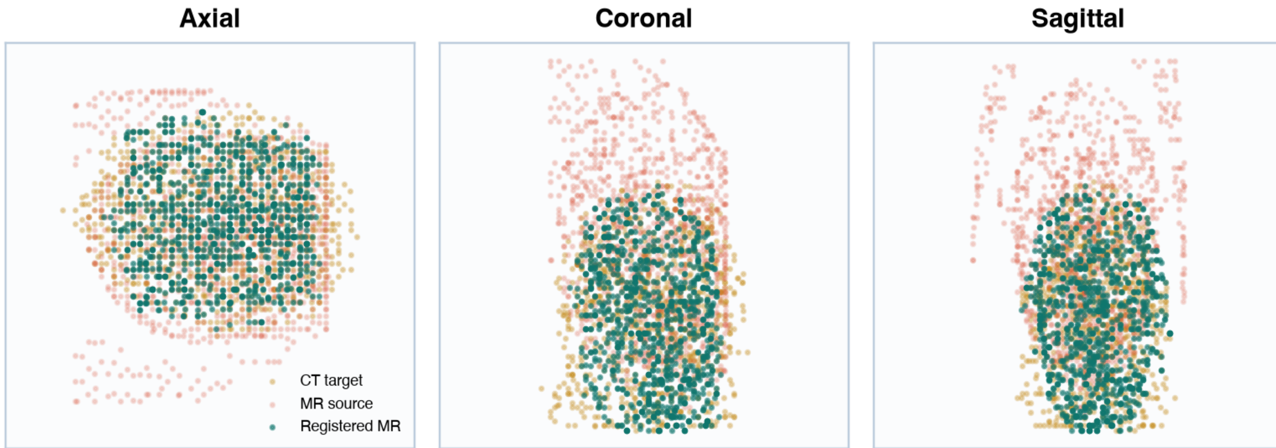

Figure 3 | Orthographic point-set projections. Downsampled source MR, target CT and registered MR surfaces for the default eight-fence configuration. Visual overlap is descriptive and does not establish anatomical landmark agreement.

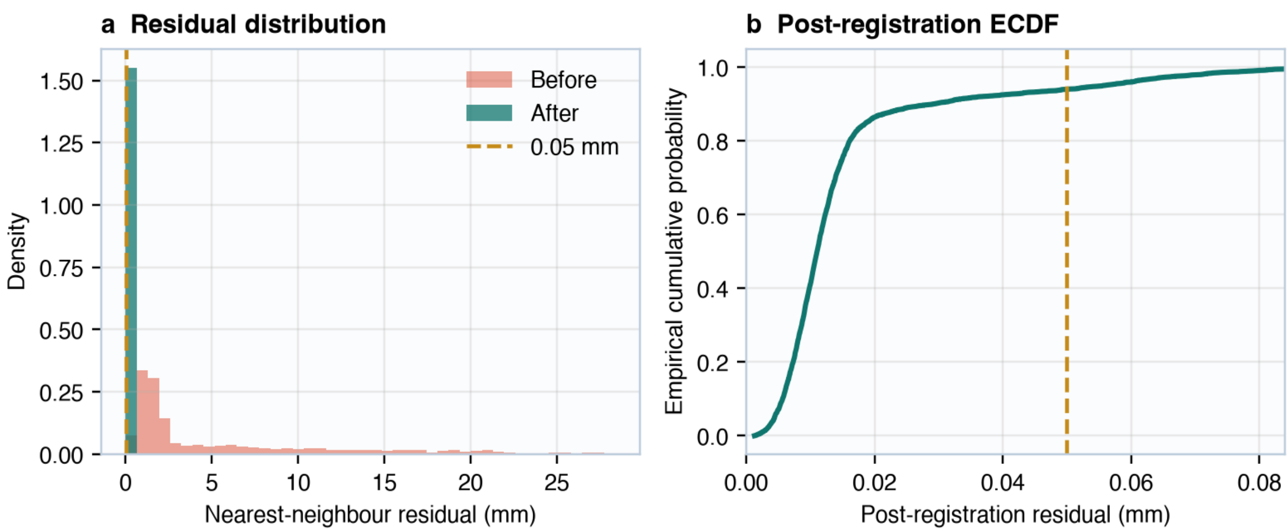

Figure 4 | Nearest-neighbour residual characteristics. (a) Residual densities before and after registration; the pre-registration x-range is truncated at its 98th percentile for visibility. (b) Empirical cumulative distribution after registration for the same 4,096 sampled points.

Discussion

Geometric fencing offers an interpretable hierarchy between a single global transform and an unconstrained per-point warp. Quantile boundaries balance target occupancy better than equal-width bins when surface density is heterogeneous, while median local displacement limits sensitivity to extreme correspondences. The approach is computationally simple: one static target k-d tree supports repeated correspondence queries, and fence updates are separable after assignment.

Limitations

This preprint reports one bundled case in an index-derived geometric frame. DICOM voxel spacing, image orientation and patient-coordinate transforms are not propagated into the evaluated surfaces; the mm label therefore reflects the application's model-space convention. No held-out landmarks, segmentations, repeated scans or clinical outcomes are available. Nearest-neighbour attraction can collapse source points onto a target surface and thereby optimize the reported residual without preserving topology or invertibility. The present implementation also lacks bending-energy, Jacobian-determinant and inverse-consistency penalties. Comparisons between fence densities reuse the same case and are sensitivity analyses, not statistical benchmarks.

Required validation

A translational study should evaluate multiple subjects in DICOM patient coordinates; use prospectively defined anatomical landmarks and segmentation overlap; report landmark TRE, Dice score, Hausdorff distance, Jacobian folding and inverse consistency; compare against rigid ICP, coherent point drift, B-spline free-form deformation and a diffeomorphic baseline; and include ablations that remove affine, fence-local and dense components. Runtime should be measured independently of I/O after warm-up on specified hardware.

Conclusion

Combinatorial geometric fencing produced a sub-0.05 mm mean nearest-neighbour surface residual in the bundled computational case while retaining a transparent sequence of global, regional and dense updates. The result supports continued algorithmic evaluation, but does not establish sub-0.05 mm anatomical TRE, scanner resolution or clinical accuracy. The principal next step is independent landmark-based validation under physical-coordinate and deformation-regularity constraints.

Code and data availability

The implementation, application endpoint and this reproducible document generator are maintained in the mersivity-2 source repository. The generator calls the local Flask test client and derives figures and table values directly from endpoint responses. Availability of imaging data remains subject to the repository's distribution and governance conditions.

References

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